

ADAPT

Where Does an AI Learn a New Rule?

A research-backed interactive laboratory for inference-time adaptation, adaptive state memory, and the BDH / BDH-CQ frontier.

The one question

When a model sees a few examples of an unfamiliar task, **where does the new rule actually live?** In the examples? In updated parameters? Or in an evolving internal state? ADAPT turns that question into an executable, inspectable experiment rather than a metaphor.

The design pressure

Most machine-learning systems are deployed with fixed parameters. Yet inference-time use often demands the opposite behavior: a system encounters a task it did not train for, sees a small number of demonstrations, and must adapt immediately. ADAPT focuses on the mechanism beneath that behavior: *what substrate carries the new task information, what does it cost, and how does it fail?*

The laboratory isolates mechanisms on identical seeded episodes. Context adjustment leaves the model unchanged and relies on the demonstrations. Parameter test-time adaptation updates weights using the demonstrations. State adaptation keeps persistent parameters fixed while writing task information into an internal state. The same outward behavior can therefore arise from different underlying changes.

MECHANISM	WHERE THE RULE LIVES	PARAMETERS	ADAPTIVE STATE
Frozen / context	Demonstrations remain in context; no persistent update.	0	0
Parameter TTA	Gradient-updated test-time parameters.	> 0	0
State adaptation	An evolving internal register is written per demonstration.	0	> 0
Learned state / TTT	A learned state mechanism adapts without persistent weight updates.	0	> 0

A controlled laboratory, not a cinematic demo

The learner-facing numbers come from the same research engine used for the experiments: a seeded task generator, adaptation strategies, telemetry, a result store, and a small standard-library HTTP server. Every live result carries provenance for the seed, split, task-generator version, model version, Git commit, configuration

hash, and schema version.

The task generator spans linear, quadratic, symbolic, compositional, and an ARC-like grid proxy. Each task has an exact hidden rule and exact ground-truth answer. The interaction pattern is deliberately simple: **predict** -> **execute** -> **compare** -> **explain**. The UI makes the scientific object itself – demonstrations, state, query, prediction, truth – the interface.

The experiment that anchors the story

State adaptation in one line

dTheta = 0 while the adaptive state changes after the demonstrations. The key question is whether that changing state is merely correlated with success or actually carries task information.

ADAPT therefore runs interference and intervention experiments. In $A \rightarrow B \rightarrow A$, a task is learned, another task is learned without resetting, and the first task is tested again. In intervention, the adapted state is perturbed, the prediction is re-evaluated, and the original state is restored. Collapse followed by recovery is stronger causal evidence than a visualization that merely moves.

EVIDENCE	RESULT	SETTING	STATUS
E1 acquisition	Frozen 0.11 -> state 1.00	200 paired linear tasks	OUR MEASUREMENT
Symbolic acquisition	Frozen 0.00 -> state 1.00	Controlled symbolic family	OUR MEASUREMENT
E4 capacity	Accuracy cliff at $d_s=4$ (0.10)	State bottleneck sweep	OUR MEASUREMENT
E5 interference	$A \rightarrow B \rightarrow A$: strong retention loss	No reset between tasks	OUR MEASUREMENT
E8 intervention	Perturb ~ 0.03 -0.10; restore 1.00	State perturb/restore	OUR MEASUREMENT
Learned recurrent	0.09 vs frozen 0.11	Meta-training budget	OUR NEGATIVE
BDH-CQ headline	29.5% pass@2 (150M)	Paper / official claim	PUBLISHED - NOT OUR RESULT

From the controlled mechanism to the frontier

The purpose of the controlled laboratory is not to claim that a toy recurrent register is a frontier architecture. It is to make the design pressure precise enough that a learner can recognize related choices in modern research.

BDH (Kosowski, Uznański, Chorowski & Stamirowska, 2025; arXiv:2509.26507) describes a post-Transformer, scale-free architecture of locally interacting neuron particles whose working memory is carried by a synaptic state updated through a Hebbian-style rule, separate from persistent parameters. ADAPT uses the same high-level envelope – demonstrations write an adaptive memory substrate and the query is solved

from that adapted substrate – while explicitly remaining a controlled analogue, not the official BDH implementation.

BDH-CQ (Engdahl, Kosowski, Chorowski & Stamirowska, 2026; arXiv:2608.09888) extends the design space by combining recurrent memory updates during inference with iterative latent computation for the query. The learner therefore encounters a second question: adaptation is not only about *where* information is written, but also about how additional inference-time computation operates on that memory.

Evidence discipline

ADAPT separates **OUR MEASUREMENT**, **OUR NEGATIVE**, **PUBLISHED**, **SYNTHETIC**, and **ILLUSTRATIVE** evidence. BDH / BDH-CQ headline numbers remain published claims unless independently reproduced. They are never merged into ADAPT experiment results.

This matters because numerical comparison becomes invalid when models, tasks, training regimes, and evaluation protocols differ. ADAPT therefore compares mechanisms and design choices when conditions are not commensurate rather than turning unrelated numbers into a leaderboard.

What the experiments show

In the current frozen evidence set, state adaptation reaches 1.00 accuracy on 200 paired unseen linear tasks where the frozen baseline is 0.11. On the controlled symbolic family, frozen is 0.00 and state adaptation reaches 1.00. A capacity bottleneck produces a sharp accuracy cliff. Shared-state accumulation can produce $A \rightarrow B \rightarrow A$ interference. Gradient-based adaptation benefits from additional test-time compute but shows diminishing returns.

The negative result matters just as much: the learned recurrent checkpoint did not acquire the target linear rule under the tested meta-training budget, remaining near the frozen baseline at 0.09 versus 0.11. The project retains this result instead of selecting only the experiments that support the preferred narrative.

Learning by interaction

The artifact is structured as a nine-stage journey: start, teach it, manipulate, stress it, break it, bridge to BDH, transfer, research lab, and results. The learner makes predictions before seeing outputs, inspects what changed, and encounters a limitation. The point is not to animate an explanation but to let the learner perform the experiment.

The participant study is intentionally treated as evidence, not decoration. The current cohort contains four distinct participants and is preliminary. Mean pre-test is 2.5/5, mean post-test is 3.25/5, and mean gain is 0.75 points. The state-versus-weights misconception fell from 3/4 to 1/4. At the same time, transfer to selecting the best strategy on an unseen task was 0/4. This pattern is useful: the current artifact appears to teach the central mechanism more readily than it teaches transfer, making transfer a measurable design target rather than a hidden failure.

Current learner-evaluation snapshot

n=4, preliminary: pre 2.5/5 -> post 3.25/5; M2 state-versus-weights 3/4 -> 1/4; transfer 0/4; BDH-CQ order quiz 2/5. These are study signals, not final learning-gain claims. The frozen protocol targets 8-15 participants.

What this project is – and is not

ADAPT is not a claim about how every frontier language model learns. It is not an ARC-AGI benchmark submission. Its grid environment is an ARC-like proxy. It is not an official BDH or BDH-CQ implementation. It is a controlled, reproducible instrument for isolating inference-time adaptation mechanisms and connecting those mechanisms to current research on adaptive memory.

That scope is the point. A falsifiable controlled system makes it possible to ask questions that are blurred inside larger models: does performance improve without changing persistent parameters? How much state capacity is needed? What happens after interference? What happens when the state is causally destroyed? Which mechanism is paying the cost – context, gradient steps, recurrent state, or latent compute?

Reproducibility and ownership

The repository packages the engine, task definitions, checkpoints, experiment configurations, expected outputs, result hashes, provenance records, accessibility and security reviews, BDH evidence ledgers, and defense notes. The frontend is deliberately inspectable: learner-facing values are either live engine results or explicitly labelled precomputed/published artifacts.

The central takeaway

Inference-time adaptation is not one mechanism. The same outward behavior can arise from different substrates, and those substrates impose different capacity, cost, retention, and interference trade-offs. ADAPT makes those differences measurable and uses BDH / BDH-CQ to place the mechanism in a contemporary research landscape.

Selected references

Kosowski et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507.

Engdahl et al. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888.

Akyürek et al. (2025). *The Surprising Effectiveness of Test-Time Training for Few-Shot Learning*. ICML 2025.

Sun et al. (2025). *Learning to (Learn at Test Time): RNNs with Expressive Hidden States*. ICML 2025.

von Oswald et al. (2023). *Transformers Learn In-Context by Gradient Descent*. ICML 2023.

Repository: github.com/Adithkp03/ADAPT